

## Introduction & conclusion script's

### the introduction part:

Good afternoon everyone. Welcome to our capstone Project presentation.

**I would like to thank and acknowledge the evaluation committee: Dr. Fahad and Dr. Hatem, as well as our supervisor, Dr. Ahmed.**

When we thought about surveillance and protecting our borders, we looked for inspiration in our own environment. In our Arab culture, the '**Eqaab**' (**Eagle**) is not just a bird; it is a symbol of **sharp vision, speed, and dominance over the sky**. An eagle sees what others miss, and it protects its territory from above.

That is exactly why we named our project '**Eqaab Drone**'. Our goal is to design an intelligent, autonomous aerial system that mimics the eagle's sharp eye to enhance security in our homeland."

Before we dive in, it is important to note that this is **Phase One** of our project. In this semester, we focused on the **System Analysis, Design, and Simulation**. We have successfully modeled the drone and tested its logic in a virtual environment. In the next phase, we **will** move to the actual hardware implementation and physical testing.

"In today's presentation, we will guide you through the following points:

- First, I will start with a general **Overview** of the technology.
- Then, my colleagues will explain the **Problem Statement** and our **Objectives**.
- After that, we will discuss the **System Design** and the **Hardware & Software Specifications** we plan to use.
- then, we will show you the **Simulation Results** we achieved.
- Finally, we'll discuss the conclusions that we noticed

### 4. Overview (Setting the Stage)

"So, let's start with an overview. **Surveillance** technology is evolving rapidly. Traditionally, security relied on fixed cameras or ground patrols. But these methods have limitations—cameras have blind spots, and humans get tired.

UAVs, or drones, are the modern solution. They provide a **flexible, bird's-eye view** that fixed systems simply cannot match. The core concept of 'Eqaab' is to combine this mobility with **Artificial Intelligence**. By using computer vision, our designed system will not just 'record' video; it will **understand** it—detecting vehicles and intruders automatically."

Now, my colleague - mohammed - will explain the specific problem we are solving in more detail."

Before we wrap up, let's talk briefly about the constraints we faced and how we addressed them

### **the conclusion part:**

To conclude our presentation for this semester: We have successfully completed the **design and simulation phase** of the Eqaab Drone. We established the system architecture, integrating the PX4 flight controller concept with an onboard AI unit. Through our Gazebo simulations, we **proved** that our logic works: the drone can fly autonomously, and the computer vision model successfully detects targets like cars and humans. We demonstrated that an AI-enabled UAV can effectively reduce blind spots and improve situational awareness.

## **2. Future Work (What is coming next)**

Looking ahead to the next semester, our roadmap is clear:

- **Hardware Assembly:** We **will** purchase the components and build the physical drone based on the specifications we presented.
- **Field Testing:** We **will** move from the simulator to the real world, testing the drone in outdoor environments.
- **Advanced Features:** We also plan to explore integrating **Thermal Cameras** for night surveillance and potentially testing multi-drone coordination.

This project is just the beginning of a robust security solution.

Thank you for your attention and guidance. We are now ready to answer your questions.